Zimmerman (ORO) computers - present & future Oct 10, 58
R. Smith (ORO future needs) (unclassified)

2 or more classes of computations

(1.) Operations Semulations

- more than simple Monte Carlo - sample histories.
simulation - but decision strategies are left open
of game

several kinds of games: (1) Tank vrsus tank operations
(majority of machine time) → (2) aircraft, surface to air missiles
(3) set of models - defense of contintal U.S.
for pre-planned attack
(4) flow of material thru army depots,
for reorders etc.

(2) "free play" semulations - no specific decisions built in
players decide as they go on.
try to follow Doctrine - but can deviate from it.
(refer to human operator or to another doctrine
same as refering to "higher hquarters"

"Gestalt" translator I/O equipment ⇒ Imp. as Technique of OR. but also to Army for auto data processing
2D graphs, tables, displays.
display of terrain - map, etc.

Intelligence information
pics, etc.
- need info. pressure of
Time deliays (several hours between echelons)

so Same equip. applies for O.R. as for Army Intelligence Display.

→ need small "breadboard" computer to try scaled down prob.
i.e. 20 $100K machines not one $2M.

Large dimensional dynamic programs non-linear.
all weapons - partial overlap.
- allocation of national resources
reserve units
etc
- need to perfect numerical techniques
(There are usually several arbitrary dimensions beyond necessary one — can put in economic features + cost
large computing quantity.
→ transfer material info. independently of logical & arith. units.

e.g. stock control problem 4 items at n depots, backorders, etc.
would require 43 tapes of 705 type of info.
- instead do one page at a time in the catalog.

→ Data Transmission Devices without human intermediaries. — 60% errors in human operation at least

— transmit at distances - 100 mi away. — C.W.S. would like to simulate these also

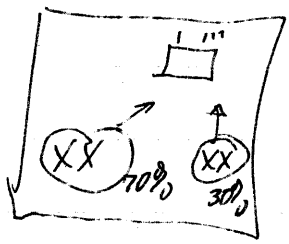
example of human intervention:

reorder time on gasolene } 5 days 30 days
keep between

but under heavy attack one might go ~~down~~ down to 0, (go against doctrine).

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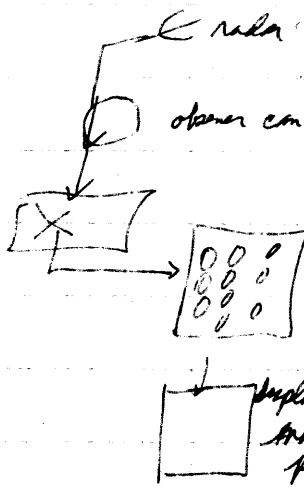
example: project onto screen "picture of war" battlefield



enemy not sure of size $a < x < b$.
attack starts.

intervention during attack — must be up to date
- change in local objectives
- change in jump-off time.

Sage:



observer makes main decisions
machines are doing more + more
eg. Vomare.

Timm: → can't state requirements of Nylon now to Army — must convince them of payoff,
— 1103 just about exhausts ORO coding capability now —
thinking smaller now, but do need to do simulation
— coding techniques, re hardware ← how should one expand support.

→ customers of future will want to do more simulation type probs. - more programs

eg. 4-5 man years of programming
+ several years.

Hebee C1

for FORTRAN type
- a large amt of data in tables
- some system for tagging 100's diff variables + to code using these tags.
- write as "check x as a string of tags"
find ∩ intersection + ∪ unions

data: logical lists 0 + 1 existence + not.
go thru lists + find classification of pattern.

eg. pick out targets for weapons.

list of "dead" enemy	11001100
list of out of range	11101101
list of tanks	011 00000100

pick out live in range, tank.

write $P_{B13} \cup R_{B13} \cup T$.

write "intersect"

eg. can a unit move in present terrain.
list of units which can make such a hill. - intersect with

table kill probabilities : fns. of many variables,
such as range, type of tanks, gun,
ammo, concealment

write Prob is fn of (· · · · ·)
Then go into table & pull out

~~Routines~~ Basic opns:
— Sorting
— Branching on basis of tests.

"Do everything in parallel"
— a single logical word incorporating state of battle field
— a logical operator · produces next step at once

|— — — — — — —| may be thousand bits long

Small battles code "Carnonet" — is unclassified,
they are willing to give talk on it in NY or Pough
anytime

Fri pm Oct 10, '58

Military Products. Talk at ORO. by John Orzano.

real time - closed loop

1. pulse magnetics
2. Transistor (RTA)
3. cryogenics

rugged characteristics. { management control / Data coordinator } +$3.8M R+D contract

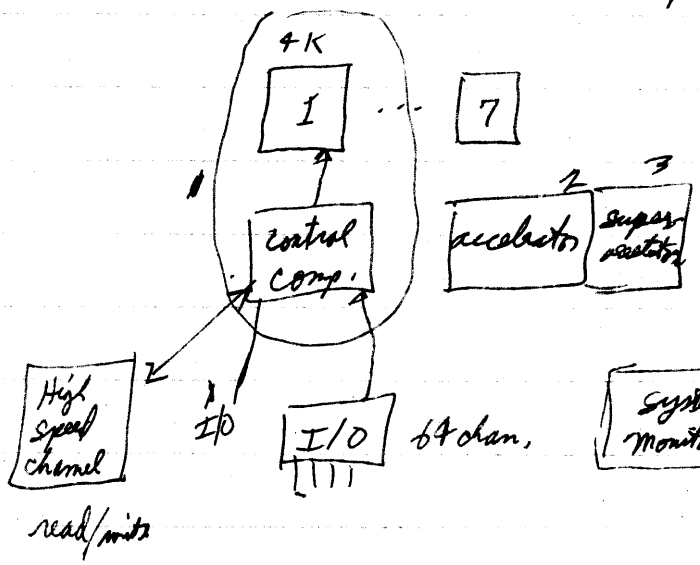
15 X 704 speed, adv. Sage., SAC Center (ITT), CPX has been proposed for Ft. Monroe

- superspeed.

① memory — proposed as "retro-fit" for RTA in 3 years.
② switching — have obtained 10 mps

500K components/craft.

1. Magnetics.



read/write

(1 memcycle)

1. 1/3 speed of 704
2. 1x speed of 704
3. A 16 μs, M 80 μs

12 to 18 mos.

question: what is relationship to Moby Dick?

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2. RTA: 1/40 size 525,000 to 600,000 instructions per second

18 mos. delivery

1 μs add
6 μs mpy
26 μs div
30 μs sq rt.

65K wd core mem,
2 banks 32K, 32K
Wd length 36 bit, 48 bit } both proposed

marginal checking - automatic

Drum 210K wd

→ 15 min per day failure rate (2 days mean error time)
1 hr per day p.m.

100 Q packs → 16 Plug-pkg units → 5 Modules → RTA machine